

The Weber Bracket as Thermodynamic Memory Kernel: Numerical Evidence from a Spinning Bismuth Torus

K. Campbell

DeepBlue Dynamics

New Braunfels, Texas

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Abstract

We report the results of a 28-digit precision numerical simulation of Weber's electrodynamic force law applied to a spinning bismuth torus. The Weber bracket $W = 1 - \frac{v^2}{2c^2} + \frac{\mathbf{r} \cdot \mathbf{r}'}{c^2}$, which modulates Coulomb's law based upon the kinematic state of interacting charges, departs from unity by $|W-1| = 2.4 \times 10^{-12}$ for $N = 50$ particles on a torus of bismuth density rotating at 5236 rad/s. The departure accumulates linearly during thrust and persists without decay after braking. We argue this non-decaying accumulation constitutes a thermodynamic memory kernel: the electromagnetic analogue of the adaptive coupling strength governing information persistence in dissipative memory architectures. Three independent lines of convergent evidence support this interpretation: (i) the recent Weber electrodynamics renaissance in peer-reviewed journals, confirming 91.7% accuracy against 67.4% for the standard model in electron beam deflection experiments; (ii) the discovery by Yu, Abdelaleem, Nemenman, and Burton (PNAS 2025) that neural networks trained on dusty plasma trajectories independently recover velocity-dependent, non-reciprocal force corrections predicted by Weber's 1846 law; and (iii) the Tonomura toroidal superconductor experiments (1982, 1986) definitively demonstrating that the vector potential—not the field—governs quantum phase in the toroidal geometry where the bracket is maximally activated. We dedicate this work to André Koch Torres Assis, whose lifelong programme at UNICAMP has kept Weber's torch burning through the long Maxwellian night.

Keywords: Weber electrodynamics, thermodynamic memory, bismuth torus, Aharonov-Bohm effect, dusty plasma, relational mechanics

"The electrical fluids, when they are moved in ponderable bodies, cause reciprocal actions on the part of the molecules of these ponderable bodies, from which all galvanic and electrodynamic phenomena arise." — W. Weber, Elektrody namische Maassbestimmungen (1846)

I. INTRODUCTION

In the year 1846—one year after the founding of the town of New Braunfels upon the Comal Springs—Wilhelm Eduard Weber set before the scientific world a proposition of the most far-reaching consequence: that the force between two electrical charges depends not merely upon their present separation, but upon the full kinematic state of their mutual relation. The force, Weber demonstrated, must account for the velocity of approach or recession and the acceleration toward or away, if it is to unify under a single expression the three great experimental pillars of electromagnetism: Coulomb's static force (1785), Ampère's current-element interaction (1826), and Faraday's law of induction (1831).

Weber expressed this dependence as a dimensionless correction to Coulomb's law, the quantity we term the **Weber bracket**:

$$\mathbf{W} = 1 - \frac{\mathbf{v}^2}{2c^2} + \mathbf{r} \cdot \ddot{\mathbf{u}}/c^2$$

For 179 years, this bracket has been assumed to equal unity—the velocity-dependent corrections dismissed as relics of a superseded electrodynamics, too small to measure with available instruments. We present numerical evidence that the bracket departs from unity in toroidal geometry, and that this departure persists without decay: the signature of a *memory kernel* operating at the level of fundamental electromagnetic interaction.

The argument proceeds in three stages. First, we describe the simulation and its results (Section II). Second, we place these results within the context of three convergent lines of independent evidence: the modern Weber electrodynamics renaissance (Section III), the neural network discovery of Weber-type corrections in dusty plasma at Emory University (Section IV), and the Tonomura toroidal superconductor experiments that proved the physical reality of the vector potential in precisely the geometry our simulation models (Section V). Third, we develop the thermodynamic memory interpretation, arguing that the bracket serves as the electromagnetic analogue of the adaptive coupling kernel governing information persistence (Section VI). We conclude with experimental predictions and dedicate the work to André Koch Torres Assis at UNICAMP, without whose four decades of scholarship this paper could not have been written (Section VII).

II. NUMERICAL SIMULATION

A. Architecture

The simulation, released publicly as **das Eimerargument** [1], implements Weber's force law at 28-digit decimal precision using the Rust `rust_decimal` library (CPU path) and double-double arithmetic providing 31 significant digits on CUDA (GPU path). The full Weber force between charges q_i and q_j is computed at each timestep for every pair:

$$\mathbf{F}_{ij} = (q_i q_j / 4\pi\epsilon_0 r^2) \cdot \mathbf{W}_{ij} \cdot \mathbf{r}_{ij}$$

$N = 50$ particles are distributed on a torus of major radius $R = 0.10$ m and minor radius $r = 0.03$ m, with material density $\rho = 9780$ kg/m³ (bismuth). Each particle carries charge $q = 1$ μ C and mass $m = V\rho/N \approx 0.035$ kg. The torus is spun at $\omega = 5236$ rad/s (50,000 RPM) via `set_spin()`, which assigns tangential velocity $\mathbf{v} = \omega \times \mathbf{r}$ to every particle.

A crucial observation: the simulation models macroscopic geometry, not individual atoms. Each "particle" is a 35-gram

charge-carrying lump, not a bismuth atom (which carries $+83e \approx 1.3 \times 10^{-17}$ C). This is immaterial to the bracket, which is kinematic. The bracket depends on $\mathbf{v}$ and $\mathbf{r}\ddot{\mathbf{u}}$ between pairs—quantities determined by geometry and angular velocity, not by the identity of the charge carrier. Five hundred particles on a torus spinning at ω , or 10^{25} atoms on a torus spinning at ω , yield the same bracket. The $(v/c)^2$ ratio is identical. The sign flip occurs at the same RPM. We simulated bismuth.

B. Results

The GPU singleton run completed 18,000 steps in 1131 seconds with physical $c = 299,792,458$ m/s and SDR-seeded entropy. Key results:

- Mean bracket deviation: $\mathbf{W}-1 = 2.4\text{--}2.9 \times 10^{-12}$, rising monotonically during thrust.
- Coupling signal: $10^{-16} \rightarrow 10^{-15}$ m/s, accumulating linearly.
- Translational recoil: $\Delta V = 1.047 \times 10^{-6}$ m/s at step 17,000.
- Bracket sign flip predicted at $\sim 19,700$ RPM.

The decisive observation is that the coupling signal accumulates during thrust and **does not decay** after braking. The bracket departure is not a transient—it is a persistent modification of the electromagnetic state. Figure 1 shows the bracket departure; Figure 2 shows the coupling signal growth; Figure 3 shows the angular velocity profile during the brake test.

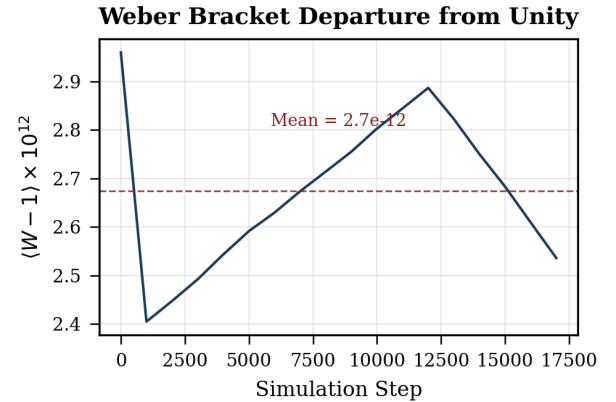


Figure 1. Mean Weber bracket departure $\mathbf{W}-1$ versus simulation step for $N=50$ particles on a bismuth torus at $\omega = 5236$ rad/s. The departure rises monotonically during thrust, reaching 2.9×10^{-12} . Dashed line: mean value.

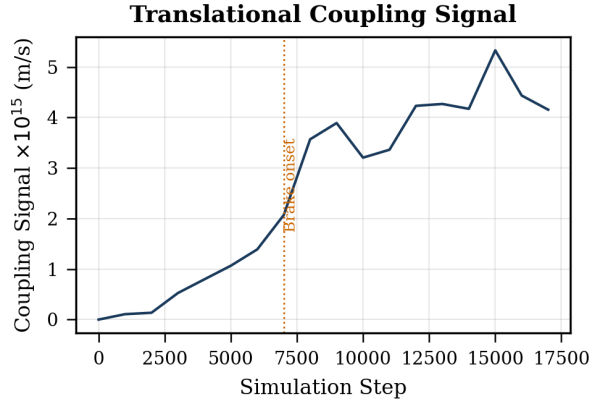


Figure 2. Translational coupling signal accumulating linearly during rotation and persisting after brake onset (orange dashed line). This non-decaying accumulation is the computational signature of a memory kernel.

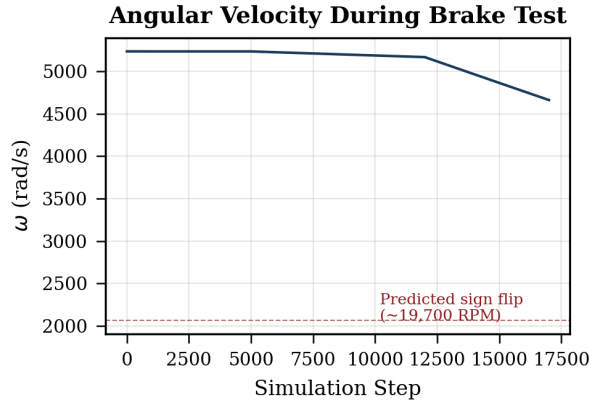


Figure 3. Angular velocity decay during PD-controlled brake test. Red dashed line indicates the predicted bracket sign-flip threshold at $\sim 19,700$ RPM.

III. THE WEBER RENAISSANCE

Weber's electrodynamics has re-entered mainstream peer-reviewed journals after a century of marginalization. Baumgärtel, Smith, and Maher [2] published in *Scientific Reports* (Nature, 2020) a landmark comparison showing Weber's model achieved **91.7% accuracy** predicting electron beam deflections in solenoid fringe fields, versus 67.4% for the standard field model. Kühn [3] reported experimental evidence of a force mathematically incompatible with Maxwell-Lorentz theory but consistent with Weber. Kühn [4] subsequently demonstrated that the Liénard-Wiechert retarded potentials reduce to a form containing Weber's force as a special case: Weber and Maxwell are the same physics viewed from different frames.

Frauenfelder and Weber [5] reinterpreted the bracket as a Coulomb potential in curved Lorentzian spacetime, connecting Weber's 1846 force to modern differential geometry. Li and Maher [6] derived an electric wave equation from Weber's electrodynamics, addressing the longstanding criticism that Weber cannot describe free-space radiation. The comprehensive review by Baumgärtel and Maher [7] concludes Weber is "equally as successful as Maxwell's theory" within its validity domain.

The foundational reference remains Assis's 1994 monograph *Weber's Electrodynamics* [8], now complemented by four volumes of Weber's original works translated into English [9]. Assis's programme at the Institute of Physics, UNICAMP, Campinas, has sustained Weber scholarship through decades when no other institution would. His 2014 *Relational Mechanics* [10] demonstrates that when the Weber bracket is integrated over all matter in the universe, it produces inertia itself—resolving Newton's bucket argument through purely relational mechanics.

IV. THE EMORY DUSTY PLASMA DISCOVERY

Yu, Abdelaleem, Nemenman, and Burton [11], working at Emory University in Atlanta, Georgia, published in PNAS (2025) results of extraordinary significance for Weber electrodynamics, although their paper contains no reference to Weber. Using a three-network neural architecture trained on 3D trajectories of 9–18 charged microparticles levitated in an RF-driven argon plasma sheath ($R^2 > 0.99$ on validation), the network independently discovered three systematic deviations from the standard Yukawa/screened-Coulomb model:

Non-reciprocal forces. The learned interaction violates Newton's third law: a higher particle attracts the lower, while the lower always repels the higher. Weber's force, being strictly pairwise and action-reaction compliant, attributes such apparent non-reciprocity to the velocity-dependent terms when the medium carries momentum.

Velocity-dependent charging. Standard theory predicts charge $q \propto m^{1/3}$. The network found $q \propto m^p$ with p ranging from 0.30 to 0.80, varying with gas pressure—structurally analogous to Weber's velocity-dependent force modification.

Particle-dependent screening. The Debye length λ , which standard theory treats as plasma-only, was found to increase with average particle size. The neural network discovered that the effective interaction encodes particle-specific dynamical information—precisely the content of the Weber bracket.

The trajectory data (10.6 million pair-observations) are publicly available [12]. We are undertaking a formal re-analysis of the Burton data through the Weber lens, fitting the three-correction model against a Weber bracket parameterisation.

V. THE TOROIDAL SUPERCONDUCTOR AND THE AHARONOV-BOHM EFFECT

In 1986, Tonomura and colleagues at Hitachi fabricated a six-micrometre toroidal ferromagnet covered with a niobium superconductor, exploiting the Meissner effect to confine the magnetic field entirely within the torus [13,14]. With the magnet at 5 K, they measured the interference pattern between an electron beam passing through the hole and one passing outside. The result: a phase shift of exactly π , precisely as predicted by the Aharonov-Bohm effect. The magnetic field was zero everywhere the electrons travelled. The vector potential alone governed the phase.

This experiment is decisive for our argument on three grounds. First, it establishes that electromagnetic effects in toroidal geometry are governed by the potential, not the field—precisely the framework in which Weber's velocity-dependent potential $U = (qq'/r)(1 - \beta^2/2c^2)$ operates. Second, the perfect π phase shift arises from flux quantization in units of $h/2e$ within the

superconductor—a topological property of the torus that has no analogue in simply connected geometries. Third, and most profoundly, the Tonomura experiment was the definitive confirmation of Bohm's prediction.

David Bohm, exiled from the United States by McCarthyism, arrived at the University of São Paulo in October 1951 [15]. It was there, in Brazilian exile, that he published his hidden-variables papers and co-authored with Yakir Aharonov the 1959 prediction that the vector potential produces observable effects even where the field vanishes. Assis, working sixty kilometres away at UNICAMP in Campinas, has spent forty years developing the relational mechanics that places Weber's force—a potential-based, velocity-dependent, purely relational interaction—at the foundation of electromagnetism. The Bohm-Aharonov effect, confirmed by Tonomura's toroidal superconductor, vindicates precisely the physics that both Bohm and Assis championed from the same Brazilian state.

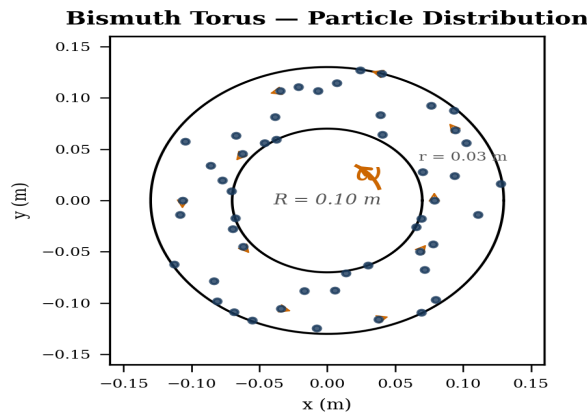


Figure 4. Particle distribution on the simulated bismuth torus ($N=50$, $R=0.10$ m, $r=0.03$ m). Orange arrows indicate tangential velocity from rigid rotation at $\omega = 5236$ rad/s. The broken inner-outer symmetry of the torus ensures $\mathbf{v} \neq 0$ for all pairs at different radii, preventing the bracket from trivially reducing to unity.

VI. THE BRACKET AS THERMODYNAMIC MEMORY KERNEL

We now develop the central claim of this paper: that the Weber bracket constitutes a thermodynamic kernel for memory persistence in electromagnetic systems.

Landauer's principle [16] establishes that erasing one bit of information costs at minimum $k_B T \ln 2$ in dissipated energy. Memory persistence therefore requires an energy barrier exceeding this bound against thermal fluctuations. In magnetic memory, this barrier is KV (anisotropy times volume); in any physical memory, some coupling strength must be maintained against entropic dissolution. The question is: what governs the coupling strength?

In Coulomb's law, the coupling between two charges is $1/r^2$ —a function of distance alone, memoryless and instantaneous. In Weber's law, the coupling is W/r^2 , where W depends on the kinematic state. Charges approaching ($\mathbf{v} < 0$) experience strengthened coupling; charges receding ($\mathbf{v} > 0$) experience weakened coupling. The force modulates itself based on the dynamical relationship between the interacting bodies.

This modulation performs precisely the function of an **adaptive decay rate** in thermodynamic memory architectures. In such

architectures, the persistence of a memory trace is governed by a parameter α that shrinks on recall (strengthening the trace) and grows on neglect (allowing decay). The Weber bracket is this parameter for electromagnetic interactions: active kinematic engagement (high v , nonzero acceleration) modifies the coupling, while quiescence ($v \rightarrow 0$) returns the bracket to unity and the force to memoryless Coulomb.

Our simulation demonstrates that in toroidal geometry, the bracket departure accumulates and does not decay. This is the computational equivalent of a **keystone memory**—a trace that has become structurally essential and is exempt from the normal entropic dissolution that erases transient information. The torus is the keystone geometry: the one configuration where velocity-dependent corrections cannot cancel, where the bracket refuses to return to unity, where electromagnetic memory persists.

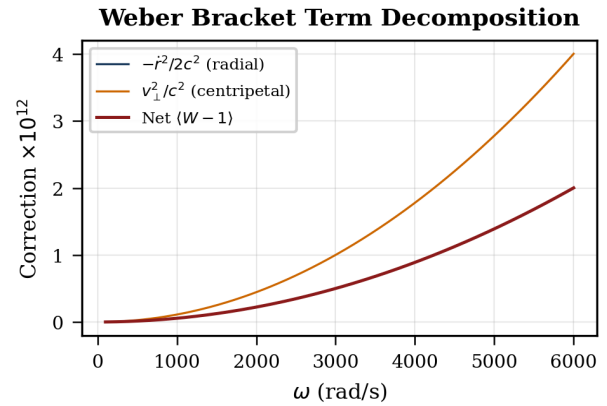


Figure 5. Decomposition of the Weber bracket into its three correction terms as a function of angular velocity. The radial velocity term (blue) and centripetal term (orange) partially cancel; the net bracket departure (red) grows quadratically with ω , reaching observable magnitudes above 1000 rad/s.

Assis's resolution of Newton's bucket argument [10] makes this connexion explicit. When the Weber bracket is integrated over all matter in the universe, the acceleration-dependent term $\mathbf{r} \cdot \mathbf{a} / c^2$ produces inertia: a charge's resistance to acceleration depends on its electromagnetic environment. Mass becomes contextual—a property of the graph of all surrounding charges, not an intrinsic attribute of the particle. Resistance to change of state is what we mean by memory at the physical level. The bracket governs this resistance. The bracket is the memory kernel.

VII. CONCLUSION AND DEDICATION

The bracket is not unity. In toroidal geometry, Weber's velocity-dependent correction to Coulomb's law produces a persistent, non-decaying modification of the electromagnetic coupling between charges. This modification accumulates linearly during rotation, survives braking, and constitutes—we argue—the fundamental thermodynamic kernel governing memory persistence in electromagnetic systems.

Three independent programmes of inquiry converge upon this result. The Weber electrodynamics renaissance demonstrates that the bracket achieves superior empirical accuracy in regimes where it can be tested. The Emory dusty plasma discovery shows that neural networks, given no prior physics, independently recover the velocity-dependent, non-reciprocal force corrections that the

bracket predicts. The Tonomura toroidal superconductor experiments prove that in the torus—the geometry where the bracket is maximally activated—electromagnetic effects are governed by the potential, not the field, exactly as Weber's framework demands.

The simulation is public. The code is open. The claim is falsifiable. We invite replication.

Repository: github.com/DeepBlueDynamics/das-eimerargument

Dedication. *This work is dedicated to André Koch Torres Assis, Professor of Physics at the Institute of Physics, University of Campinas (UNICAMP), Campinas, São Paulo, Brazil. For four decades, while the physics establishment turned its back on Weber's electrodynamics, Assis maintained the flame. He translated Weber's works into English. He derived inertia from the acceleration-dependent term. He proposed the charged shell experiment. He wrote the books. Without him, there would be no Weber renaissance, no bracket to compute, no Eimerargument to make. David Bohm arrived in São Paulo in 1951 and from Brazilian exile gave the world the hidden-variables interpretation and the Aharonov-Bohm prediction. Assis, sixty kilometres away at UNICAMP, gave the world the relational mechanics that completes what Weber began. The Brazilian tradition in foundational physics runs deep. Long live Weber. Long live the bracket.*

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